

```

%root_path = fileparts(mfilename("fullpath"));
root_path = pwd;
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit200', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei200meV.mat'));
    cuts2fit200 = ld.cuts2fit200;
end
if exist('used_ds200', 'var')
    used_ds = used_ds200;
else
    used_ds = containers.Map()
    sel_info = struct('cuts', [], 'en_range', [], 'fname', '', 'bg_par', []);
    %
    sel_info.cuts = cuts2fit200.cuts_dir001FFfit(1);
    sel_info.en_range = 30:10:110;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200ref001off110_2D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei200Br110dir001')=sel_info;
    %
    sel_info.cuts = cuts2fit200.cuts_dir001FFfit(2);
    sel_info.en_range = 30:10:130;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200ref001off200_2D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir001')=sel_info;
    %
    sel_info.cuts = {cuts2fit200.cut200_sel};
    sel_info.en_range = 40:10:140;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200Br200dir010_2peaks_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir010_2peaks')=sel_info;
    %
    sel_info.cuts = {cut(cuts2fit200.cut200_sel, [0,0.02,1], [])};
    sel_info.en_range = 40:10:140;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200Br200dir010_1peak_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir010_1peak')=sel_info;
    %
    sel_info.cuts = {cut(cuts2fit200.cut110_sel, [0,0.02,2], [])};
    sel_info.en_range = 30:10:110;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200Br110dir010_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br110dir010')=sel_info;

    %-----
    sel_info.cuts = cuts2fit200.cuts_dir001FFfit(1);
    sel_info.en_range = 30:10:110;

```

```

sel_info.cut_dim = 1;
sel_info.fname='EnFit_Ei200ref001off110_1D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei200Br110dir001_1D')=sel_info;
%
sel_info.cuts = cuts2fit200.cuts_dir001FFfit(2);
sel_info.en_range = 30:10:130;
sel_info.cut_dim = 1;
sel_info.fname='EnFit_Ei200ref001off200_1D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei200Br200dir001_1D')=sel_info;

used_ds200 = used_ds;
end

```

```
used_ds =
```

Map with properties:

```

    Count: 0
    KeyType: char
    ValueType: any
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 4690699 pixels -- processing data... -----> retained 4604422 pixels
-----
Number of points in input object: 8259853
  Fraction of object selected: 56.79% (= 4690699 points) ; Access speed: 1.27GB/sec
  Fraction of object retained: 55.74% (= 4604422 points)

**** Total time in cut(sqw) : 0.8sec Including:
    Data access time fraction: 32.9%; Transf. time frac.: 41.6%; Resulting pix preparation: 25.5%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 9054566 pixels -- processing data... -----> retained 8994792 pixels
-----
Number of points in input object: 10741088
  Fraction of object selected: 84.30% (= 9054566 points) ; Access speed: 1.44GB/sec
  Fraction of object retained: 83.74% (= 8994792 points)

**** Total time in cut(sqw) : 1.2sec Including:
    Data access time fraction: 34.2%; Transf. time frac.: 41.5%; Resulting pix preparation: 24.3%
-----

```

```

%cuts_list = cuts2fit200.cutsS_list();
%cuts_list = {cuts2fit200.cut200_200sel};
%cuts_list = {cuts2fit200.cut110_sel};

select= 'Ei200Br200dir001_1D';
sel_info = used_ds(select);

cuts_list = sel_info.cuts

```

```
cuts_list = 1x1 cell array
{1x1 sqw}
```

```
mi = maps_instrument(200,600,'S');
```

```

sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```

```

dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning
%cut_en = 30:10:120;
%cut_en = 40:10:150;
%cut_en = 145;
cut_en = sel_info.en_range;
cut_dimensions = sel_info.cut_dim;

bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;

```

```

[fit_res_rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);

```

```

*****
**** En = 40±5
*****

```

```

*****
**** En = 50±5
*****

```

```

*****
**** En = 60±5
*****

```

```

*****
**** En = 70±5
*****

```

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.2258

```
*****
**** En = 80±5
*****
```

```
*****
**** En = 90±5
*****
```

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.2258

```
*****
**** En = 100±5
*****
```

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.2258

```
*****
**** En = 110±5
*****
```

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.2258

```
*****
**** En = 120±5
*****
```

```
*****
**** En = 130±5
*****
```

```
*****
**** En = 140±5
*****
```

Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
Amplitude,
e:\SHARE\FE\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits\EnFit_Ei200Br200dir010_1peak_2D_dirGH_dE20_fitSlopeDE.mat
Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
J0,
e:\SHARE\FE\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits\EnFit_Ei200Br200dir010_1peak_2D_dirGH_dE20_fitSlopeDE.mat

Warning: Error in state of SceneNode.

String scalar or character vector must have valid interpreter syntax:

Gamma,

e:\SHARE\FE\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits\EnFit_Ei200Br200dir010_1peak_2D_dirGH_dE20_fitSlopeDE.mat

```
beep  
beep  
beep
```

```
bg_data = extract_bg_par(fit_res_rec.all_fit_par,[1,2]);  
% original slope parameters  
pd(bg_data(1));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);  
keep_figure;
```

```
pd(bg_data(2));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);  
keep_figure
```

```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_  
figure;  
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_  
figure
```

```
fit_resEi200 = fit_res_rec;  
save(res_name,"fit_resEi200")
```